

Question	Answer	Mark	Mark scheme	Additional guidance
20	$h = \frac{120}{\sqrt{t}}$	P1	for setting up a proportional relationship between h and p , eg $h \propto \frac{1}{p}$ or $h = \frac{k}{p}$ OR a proportional relationship between p and t , eg $p \propto \sqrt{t}$ or $p = K\sqrt{t}$	Condone the use of ‘ α ’ instead of ‘=’ for the first two P marks Relationship may be implied by substitution
		P1	for process to substitute at least 2 values, eg $10 = \frac{k}{6}$ ($k = 60$) or $6 = K\sqrt{144}$ ($K = 0.5$)	
		P1	for full process leading to $h = \frac{"60"}{p}$ oe and $p = "0.5"\sqrt{t}$ oe	Both constants must come from a correct process
		A1	$h = \frac{120}{\sqrt{t}}$ oe eg $h = \frac{120\sqrt{t}}{t}$ or $h = \frac{60}{0.5\sqrt{t}}$	Formula for h in terms of t Does not need to be in simplest form